Information-Guided View Selection Model for Intraprocedural M-TEER Using Coaptation Geometry and Residual Mitral Regurgitation Uncertainty

Prihantini
Master Student, Department of Financial Engineering, WorldQuant University, Washington. D.C, USA

Abstract

Mitral transcatheter edge-to-edge repair (M-TEER) depends on sequential echocardiographic assessment of coaptation geometry, device–leaflet interaction, residual mitral regurgitation (MR), and transmitral hemodynamics, yet no quantitative framework currently determines which acquisition should be prioritized when several clinically relevant quantities remain uncertain. We developed a literature-parameterized computational model that formulates intraprocedural view selection as a sequential information-allocation problem. A four-dimensional latent state represented coaptation gap, coaptation length, residual three-dimensional vena contracta area (3D-VCA), and transmitral mean gradient. Seven candidate echocardiographic acquisitions were represented by view-specific observation matrices. Gaussian belief states were updated using Kalman-form posterior covariance recursion, while multivariate differential entropy quantified expected information gain. A phase-aware utility combined information gain, clinically weighted uncertainty reduction, procedural relevance, and acquisition penalty across targeting, alignment, grasping, and result-assessment phases. Stopping was governed by prespecified multidomain uncertainty guardrails. Clinical literature provided contextual anchors, whereas prior uncertainty, observation noise, utility weights, acquisition penalties, and stopping tolerances remained explicitly provisional. Computational evaluation included a reference sequence, 600 synthetic scenarios for structural stress testing, five policy comparators, 1,200 model-decision states for internal AIC/BIC mechanism recovery, 500 Latin-hypercube parameter sets with PRCC analysis, structural sensitivity scenarios, and covariance-map stability analysis. In the reference scenario, the phase-aware policy selected 3D MPR, 3D color VCA, and continuous-wave Doppler, satisfying the stopping rule after three acquisitions with cumulative information gain of 5.432 nats. Across 600 synthetic scenarios, the phase-aware policy required a median of three acquisitions and satisfied the model stopping rule in all simulations, compared with six acquisitions for the prespecified fixed sequence. The entropy-only policy also required a median of three acquisitions, indicating that the current experiments do not establish superiority over every adaptive alternative. In the separate internal mechanism-recovery experiment, the phase-aware representation yielded AIC 715.0, BIC 760.8, and internal action agreement of 0.910. These metrics reflect recovery of a model-generated decision mechanism rather than clinical predictive performance. Global sensitivity of terminal entropy was greatest for the coaptation-length stopping tolerance (PRCC 0.561), and the complementary 3D-MPR to 3D-color-VCA to continuous-wave-Doppler covariance cycle had a spectral radius of 0.571. This study provides a reproducible mathematical proof-of-concept for selecting the next informative M-TEER acquisition according to unresolved anatomical and hemodynamic uncertainty. All numerical findings are model-based results from literature-anchored assumptions and synthetic computational experiments, not patient-level observations. Clinical calibration and external validation are required before the model can support patient-level recommendations.

Keywords: Coaptation Geometry, Information Gain, Intraprocedural Echocardiography, M-TEER, Residual Mitral Regurgitation, Sequential Decision Modeling

1. INTRODUCTION

Mitral transcatheter edge-to-edge repair (M-TEER) is fundamentally image-mediated. Intraprocedural transesophageal echocardiography is used not only to visualize mitral anatomy but also to guide device trajectory, assess orientation relative to the line of coaptation, evaluate leaflet capture, quantify residual mitral regurgitation (MR), and identify transmitral obstruction before procedural completion. Contemporary guidance therefore treats two-dimensional imaging, biplane imaging, three-dimensional (3D) en-face visualization, multiplanar reconstruction (MPR), color Doppler, and continuous-wave Doppler as complementary sources of procedural information rather than interchangeable views (Fan et al., 2022; Little et al., 2026). The most recent American Society of Echocardiography guidance formalizes the imaging content required at successive procedural steps while explicitly recognizing that no standard algorithm currently directs the interventional echocardiographer through a successful M-TEER procedure (Little et al., 2026).

The anatomical and hemodynamic quantities assessed during this workflow also have established clinical relevance. Historical EVEREST-derived selection frameworks emphasized coaptation geometry, including coaptation length and depth, flail gap and width, leaflet dimensions, and mitral valve area, although these historical criteria should not be interpreted as contemporary absolute eligibility thresholds. More recent evidence reinforces the importance of coaptation geometry in contemporary TEER populations. In a 2,140-patient cohort of functional MR, 219 patients (10%) had a visible leaflet coaptation gap, and its presence was associated with less complete MR reduction after TEER. These observations support coaptation gap and coaptation length as clinically interpretable anatomical components of an intraprocedural uncertainty state rather than arbitrary mathematical variables.

Residual MR and post-repair hemodynamics provide complementary information after device deployment. The MITRA-PRO registry evaluated 823 patients with intraprocedural residual MR quantified using 3D vena contracta area (3D-VCA) and demonstrated an association between increasing residual 3D-VCA and higher one-year mortality; Rottländer et al. further reported clinically distinct residual-MR strata based on 3D-VCA (Rottländer et al., 2025). Similarly, large observational datasets have emphasized the prognostic importance of residual MR following TEER. In the OCEAN-Mitral Registry, Kubo et al. (2023) examined one-year outcomes according to residual MR after MitraClip treatment, further supporting residual regurgitation as an important post-procedural state rather than merely an image-quality endpoint. Together with transmitral gradient assessment, these findings motivate a multidomain representation in which anatomical adequacy, residual regurgitation, and post-repair hemodynamics remain distinct but jointly relevant sources of uncertainty.

Existing literature therefore provides substantial guidance on what should be imaged, which anatomical and hemodynamic quantities matter, and how individual views contribute during M-TEER. What remains less formalized is the sequential allocation of those acquisitions when uncertainty is not distributed equally across procedural domains. Once coaptation geometry has been sufficiently resolved, for example, another geometry-focused acquisition may provide less incremental information than an acquisition directed toward residual MR or transmitral gradient. Conversely, early in the procedure, detailed residual-MR assessment may have limited decision value while device position and leaflet interaction remain unresolved. The practical question is consequently not simply whether a particular echocardiographic view is useful, but which available acquisition has the greatest marginal information value at the current procedural state. The absence of a standardized directing algorithm identified by Little

et al. (2026) provides a specific methodological gap for formal sequential decision analysis rather than evidence that existing echocardiographic guidance is inadequate.

We therefore formulated intraprocedural M-TEER imaging as a partially observed sequential decision problem in which the next acquisition is selected according to its expected reduction of unresolved anatomical and hemodynamic uncertainty relative to its acquisition burden. The framework represents coaptation gap, coaptation length, residual 3D-VCA, and transmitral mean gradient as a latent state; updates uncertainty after each acquisition; and assigns phase-dependent value to candidate views according to the procedural question that remains unresolved. A value-of-information interpretation, conceptually related to decision analysis and financial engineering, is used to quantify whether the expected information obtained from an additional acquisition justifies its modelled cost. The financial analogy is limited to information allocation under uncertainty: clinical relevance remains attached to anatomically and hemodynamically meaningful variables, and no financial quantity is mapped onto patient physiology.

The objective of this study was to establish and computationally stress-test this information-guided architecture as a reproducible mathematical proof-of-concept. Specifically, we examined whether complementary acquisitions contract multidomain uncertainty, whether view preference changes with procedural phase, whether an explicit stopping rule can terminate further acquisition once prespecified uncertainty guardrails are satisfied, and how sensitive these behaviors are to provisional structural assumptions. The framework is not presented as a clinically validated view-selection algorithm or as a predictor of patient outcomes. Rather, it defines an auditable mathematical architecture whose observation models, noise parameters, phase weights, acquisition costs, and stopping criteria can subsequently be estimated, challenged, or replaced using time-stamped patient-level M-TEER imaging data.

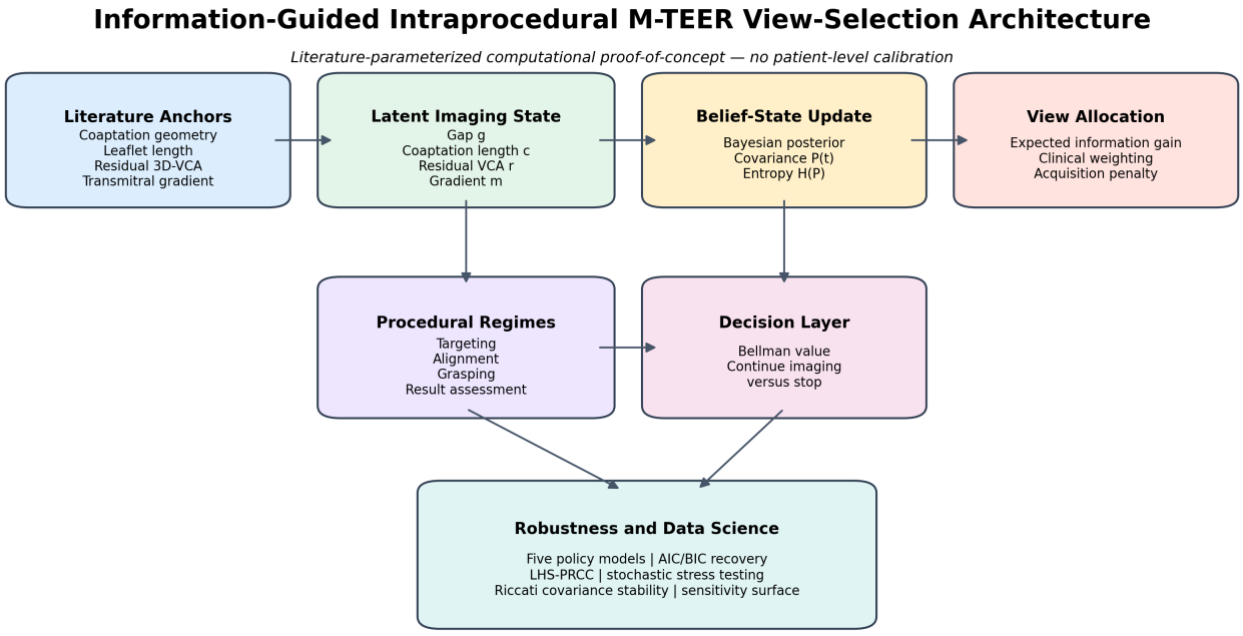


Figure 1. *Hybrid information-guided architecture linking literature anchors, latent anatomical/hemodynamic states, Bayesian belief updating, phase-aware view allocation, and stopping decisions.*

2. METHODS

a. Study Design and Scope

This study was designed as a computational proof-of-concept incorporating deterministic covariance updates and stochastic simulation experiments for information-guided echocardiographic view selection during M-TEER. Clinical literature was used to define clinically interpretable state variables and contextual anchor values, whereas parameters governing epistemic uncertainty, observation noise, acquisition burden, phase weighting, and stopping behavior remained explicitly provisional. No patient-level echocardiographic images or clinical outcome data were used to fit or validate the proposed model. Accordingly, the computational experiments were designed to evaluate mathematical behavior, internal consistency, sensitivity to assumptions, and reproducibility rather than clinical predictive performance.

b. Literature-Derived Sources

The sources were selected to preserve correspondence between the mathematical state and quantities routinely considered during M-TEER. Historical EVEREST-derived criteria included coaptation length ≥2 mm, coaptation depth <11 mm, flail gap <10 mm, flail width <15 mm, and mitral valve area around 4 cm². Leaflet length around 10 mm was used as an anatomical reference rather than an eligibility threshold. These historical values were retained as contextual anchors and were not treated as contemporary absolute eligibility criteria. Residual MR was represented using residual 3D vena contracta area (3D-VCA), with 0.10 and 0.30 cm² retained as literature-informed reference strata based on contemporary intraprocedural residual-MR evidence, including the MITRA-PRO analysis (Rottländer et al., 2025). Transmitral mean gradient was represented with 5 mmHg as a clinically familiar post-TEER reference boundary, consistent with contemporary studies evaluating residual MR and transmitral gradient after M-TEER. These quantities were used to establish clinically interpretable scales and were not fitted to the synthetic experiments.

c. Latent Anatomical Hemodynamic State

The intraprocedural state at acquisition step t was represented by the four-dimensional latent vector $x_t = [g_t, c_t, r_t, m_t]$, where g_t denotes coaptation gap (mm), c_t coaptation length (mm), r_t residual 3D-VCA (cm²), and m_t transmitral mean gradient (mmHg). These variables were selected because they represent complementary domains of procedural uncertainty: coaptation geometry, leaflet interaction, residual regurgitation, and post-repair hemodynamics. Contemporary procedural imaging literature supports assessment of these domains using complementary two-dimensional, biplane, color-Doppler, and spectral-Doppler acquisitions (Fan et al., 2022 and Little et al., 2026).

d. Gaussian Belief-State Representation

Because the true anatomical hemodynamic state is not assumed to be completely known from any single acquisition, the model maintains a Gaussian belief state $b_t = N(\mu_t, P_t)$, where μ_t represents the current estimate of the latent state and P_t represents its posterior covariance. The covariance matrix therefore quantifies the modelled uncertainty remaining in each state variable and their covariance structure. Between acquisitions, a process covariance matrix Q permits small physiological, device-related, and image-state perturbations. Both the initial covariance and Q were treated as provisional modelling assumptions rather than empirically estimated patient-level parameters.

e. Candidate Echocardiographic Acquisitions

Seven candidate acquisitions were included:

1) intercommissural view
2) LVOT long-axis view
3) biplane X-plane imaging
4) 3D en-face imaging
5) 3D multiplanar reconstruction (3D MPR)
6) 3D color VCA, and
7) continuous-wave (CW) Doppler.

Each candidate view v was represented by an observation matrix H_v, defining which components of the latent state were informed by that acquisition, and an observation-noise covariance matrix R_v, representing provisional measurement uncertainty. Intercommissural and LVOT long-axis views primarily informed geometric states; biplane X-plane, 3D en-face imaging, and 3D MPR provided complementary information on gap and coaptation geometry, 3D color VCA primarily informed residual MR, and CW Doppler informed transmitral gradient. These functional assignments were based on contemporary procedural echocardiographic practice described by Fan et al. (2022) and Little et al. (2026). The numerical entries of H_v and R_v were modelling assumptions and were not estimated from patient images.

f. Sequential Bayesian Updating and Information Gain

Following each acquisition, the predicted belief state was updated using a linear-Gaussian Kalman-form recursion. The posterior covariance was obtained through a Riccati-type covariance update, such that an informative observation reduced uncertainty in the state dimensions observed by the selected acquisition. Expected information gain was quantified as the reduction in multivariate Gaussian differential entropy between the predicted and posterior belief states. This formulation allowed candidate views to be compared according to their expected marginal reduction in uncertainty rather than according to image type alone. The resulting information gain was interpreted exclusively as a mathematical quantity within the specified observation model and not as a direct measure of diagnostic accuracy or real-world echocardiographic precision.

g. Phase Aware Acquisition Utility

Information gain alone may favor an acquisition that substantially reduces mathematical uncertainty while providing limited relevance to the current procedural task. A phase-aware acquisition utility was therefore defined to combine four components:

1) expected multivariate information gain,
2) clinically weighted normalized posterior-SD reduction,
3) phase-specific relevance, and
4) a dimensionless acquisition-cost penalty.

Four latent procedural regimes were considered: targeting, alignment, grasping, and result assessment. These regimes approximate the stepwise M-TEER workflow described in contemporary procedural imaging guidance (Fan et al., 2022 and Little et al., 2026). For belief state b and procedural regime z, the policy $\pi(b, z)$ selected the candidate acquisition with the greatest expected utility. The phase representation was prespecified and was not learned from clinical procedural sequences.

h. Uncertainty Based Stopping Rule

Acquisition was allowed to stop only after a minimum of three views had been selected and all four posterior standard deviations simultaneously satisfied the following prespecified tolerances: coaptation gap ≤0.65 mm, coaptation length ≤0.50 mm, residual 3D-VCA ≤0.055 cm^2, and transmitral mean gradient ≤0.55 mmHg. These thresholds were defined solely as computational stopping guardrails. They should not be interpreted as recommended echocardiographic precision targets, measurement-error limits, or clinically validated procedural thresholds. The sequential acquisition problem was additionally expressed through a finite-horizon Bellman interpretation in which continued imaging was justified only when the expected value of an additional acquisition exceeded its modelled burden under the specified utility structure.

i. Mathematical Consistency and Covariance Stability

Internal mathematical consistency was evaluated before comparative policy analysis. Covariance matrices were assessed for positive-semidefinite behavior, valid log determinants, bounded posterior-SD trajectories, and monotonic information reduction following observation. To evaluate repeated acquisition behavior, covariance maps were iterated toward stationary states for individual views and complementary acquisition cycles. Because a symmetric 4 × 4 covariance matrix contains ten independent elements, finite-difference Jacobians were constructed over these ten covariance components. Spectral radii were then calculated to evaluate local contraction of the corresponding Riccati-type covariance maps. A spectral radius below one was interpreted as local contraction of the modelled covariance state under the corresponding repeated acquisition cycle, not as evidence of clinical stability or patient-level outcome improvement.

j. Synthetic Policy Stress Testing

Policy robustness was examined using 600 literature-consistent synthetic truth scenarios. These scenarios were computational states generated for structural stress testing and were not simulated patients, synthetic clinical records, or estimates of a clinical population. Five decision policies were compared: prespecified fixed sequence, variance-greedy selection, entropy-only selection, clinical weighting without procedural phase; and full phase-aware selection. The comparison examined acquisition burden and terminal uncertainty under the same computational environment. No clinical discrimination or outcome prediction was inferred from these policy comparisons.

k. Internal AIC/BIC Mechanism-Recovery Experiment

A separate internal experiment evaluated whether simplified statistical representations could recover the stochastic action-selection mechanism specified by the full computational framework. A total of 1,200 model-generated decision states were generated, and stochastic action labels were produced from the specified action mechanism. Competing representations were compared using the Akaike information criterion (AIC), Bayesian information criterion (BIC), and internal action agreement. The evaluated representations were entropy-only, geometry-weighted, clinical weighting without phase, cost-only, and the full phase-aware representation. Because both the decision states and action labels were generated within the computational framework, this analysis was interpreted strictly as an internal mechanism-recovery experiment. AIC, BIC, and action agreement from this experiment do not measure clinical discrimination, external predictive accuracy, or superiority in patient care.

l. Global and Local Sensitivity Analyses

Global parameter sensitivity was evaluated using 500 Latin-hypercube parameter sets spanning provisional structural, uncertainty, cost, and stopping parameters. Partial rank correlation coefficients (PRCCs) were calculated after ranked residualization to quantify monotonic associations between individual parameters and model outputs, including terminal posterior entropy. Local sensitivity was assessed using ±1% perturbations around selected reference parameters, with elasticity used to quantify the proportional response of model outputs to small parameter changes. Both analyses assessed sensitivity to modelling assumptions rather than biological or epidemiological variability in a patient population.

m. Computational Reproducibility and Statistical Scope

All computational analyses used a fixed random seed of 20260903. Model-generated outputs included the reference acquisition sequence, policy stress tests, internal mechanism-recovery experiment, Latin-hypercube samples, PRCC analysis, structural scenarios, and covariance-stability calculations. Because no raw patient-level echocardiographic imaging or clinical outcome dataset was analyzed, ROC curves, clinical calibration curves, clinical sensitivity or specificity, hazard ratios, and clinical-effect p values were not calculated for the proposed view-selection policy. Numerical outputs from the present study therefore quantify behavior of the specified mathematical framework and should not be interpreted as estimates of clinical performance.

n. Mathematical Model Equations

Latent anatomical-hemodynamic state: $x_t = [g_t, c_t, r_t, m_t]^T$

Gaussian belief state: $b_t = N(\mu_t, P_t)$

Prediction covariance: $P_t^{-1} = P_{\{t-1\}} + Q$

View-specific observation model: $y_{t(v)} = H_v\, x_t + \varepsilon_v,\ \ \varepsilon_v \sim N(0, R_v)$

Kalman gain matrix: $K_t = P_t^{-1}\, H_v^T\, [H_v\, P_t^{-1}\, H_v^T + R_v]^{-1}$

Posterior state mean: $\mu_t = \mu_t^{-1} + K_t\, [y_t - H_v\, \mu_t^{-1}]$

Posterior covariance: $P_t = [I - K_t\, H_v] P_t^{-1}$

Multivariate differential entropy: $H(P_t) = \frac{1}{2}\, \{d[1 + log(2\pi)] + \log\det(P_t)\}$

Expected information gain: $IG(v|b_t) = H(P_t^{-1}) - H(P_{t(v)})$

Clinically weighted uncertainty reduction: $\Delta U_v = \Sigma_j\, w_j \left[\frac{\sqrt{\mathrm{P}_{\{jj\}}^{-1}}}{\mathrm{s_j}} - \frac{\sqrt{\mathrm{P}_{\{jj\}}^{-1}(v)}}{\mathrm{s_j}}\right]$

Phase-conditioned acquisition utility: $A(v|b_t, z_t) = \varphi_v(z_t)[0.58\, IG(v|b_t) + 0.42\, \Delta U_v] - \lambda\, c_v$

Sequential view policy: $\pi*(b_t, z_t) = \arg\max_{\mathrm{v}} A(v|b_t, z_t)$

Bellman decision representation: $V_t(b, z) = \max\left\{U_{stop}(b, z), \max_{\mathrm{v}}\left[-\lambda c_v + E\{V_{\{t+1\}}(b^v, z^v)\}\right]\right\}$

Information stopping time: $\tau* = \inf\{t \geq 3 : \sqrt{(P_{\{jj,t\}})} \leq \delta_j\ for\ all\ j\}$

Latent procedural regime: $z_t \in \{Targeting, Alignment, Grasping, Result\ assessment\}$

State-transition law: $\Pr(z_{\{t+1\}} = k \,|z_t = j, b_t, v_t) = T_{\{jk\}}(b_t, v_t)$

Riccati covariance map: $F_{v(P)} = P + Q - (P+Q)H_v^T[H_v(P+Q)H_v^T + R_v]^{\{-1\}}H_v(P+Q)$

Complementary-cycle stability criterion: $\rho\{D(F_{v_k} \circ \ldots \circ F_{\{v_1\}})(P*)\} < 1$

Akaike information criterion: $AIC = 2k - 2\log L$

Bayesian information criterion: $BIC = k\log n - 2\log L$

Partial rank correlation: $PRCC(\theta_j, Y) = Corr(rank - residual\ \theta_j, rank - residual\ Y)$

Note:

- An identity state-transition model was assumed between consecutive acquisitions.
- The coefficients 0.58 and 0.42 are provisional reference weights assigned to information gain and clinically weighted uncertainty reduction, respectively. They were not estimated from patient data, and their influence was examined through sensitivity analyses.
- s_j denotes the reference scaling factor for latent-state dimension j, and w_j denotes its provisional clinical importance weight.
- L denotes the likelihood of the model-generated stochastic action labels, k is the number of estimated parameters in each candidate representation, and $n = 1{,}200$ is the number of model-decision states. These criteria evaluate internal mechanism recovery rather than clinical predictive performance.

Table 1. Selected literature-Derived Clinical Anchors used to Define Variables

Clinical anchor	Value	Unit	Role in the model
EVEREST favorable coaptation length	2	mm	Minimum historical coaptation-length anchor for geometry
EVEREST unfavorable coaptation depth	11	mm	Historical leaflet-to-annulus geometric complexity anchor
EVEREST flail gap boundary	10	mm	Historical degenerative-MR anatomical complexity anchor
EVEREST flail width boundary	15	mm	Historical degenerative-MR anatomical complexity anchor
Multiplanar leaflet length favorable anchor	10	mm	Leaflet tissue/graspability anchor
Mitral valve area historical anchor	4	cm^2	Historical stenosis-risk anchor
Transmitral mean-gradient reference boundary	5	mmHg	Common post-TEER hemodynamic stratification anchor
Residual 3D-VCA favorable boundary	0.1	cm^2	Intraprocedural residual-MR quantitative anchor
Residual 3D-VCA relevant MR boundary	0.3	cm^2	Intraprocedural residual-MR quantitative anchor
Coaptation-gap prevalence in FMR TEER cohort	10	%	Prevalence/context anchor, not a model probability
AFMR 3D-TEE procedural-feature cohort	32	patients	Outcome-context anchor
AFMR 3D-TEE gradient >5 cohort	35	of 130	Outcome-context anchor

The selected literature-derived clinical anchors in Table 1 used to define clinically interpretable anatomical and hemodynamic variables, including historical EVEREST geometric criteria, leaflet length, mitral valve area, transmitral gradient, residual 3D-VCA, and contextual M-TEER cohort characteristics; these values provide clinical context and scaling rather than fitted patient-level parameters, and historical thresholds are not interpreted as contemporary absolute eligibility criteria.

Table 2. Provisional Structural Parameters and Decision Guardrails

Parameter	Value	Unit
Prior mean coaptation gap	6	mm
Prior mean coaptation length	3.6	mm
Prior mean residual 3D-VCA	0.22	cm^2
Prior mean transmitral gradient	3.8	mmHg
Prior SD coaptation gap	2.6	mm
Prior SD coaptation length	1.4	mm
Prior SD residual 3D-VCA	0.14	cm^2
Prior SD transmitral gradient	1.5	mmHg
Stopping SD coaptation gap	0.65	mm
Stopping SD coaptation length	0.50	mm
Stopping SD residual 3D-VCA	0.055	cm^2
Stopping SD transmitral gradient	0.55	mmHg
Acquisition cost weight	0.18	Dimensionless

The provisional structural parameters and decision guardrails in Table 2 used to initialize and operate the computational framework, including prior means and standard deviations for coaptation gap, coaptation

length, residual 3D-VCA, and transmitral gradient, together with posterior-SD stopping thresholds and the acquisition-cost weight; these quantities are modelling assumptions rather than clinically validated estimates and require calibration using longitudinal intraprocedural M-TEER imaging before patient-level interpretation.

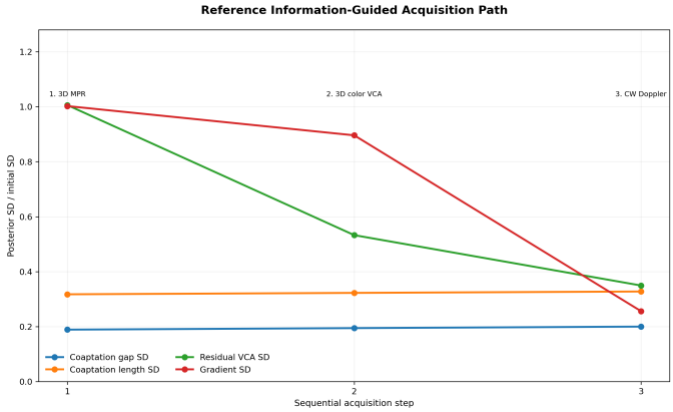


Figure 2. *Normalized posterior uncertainty across the reference acquisition sequence. Values are model-based.*

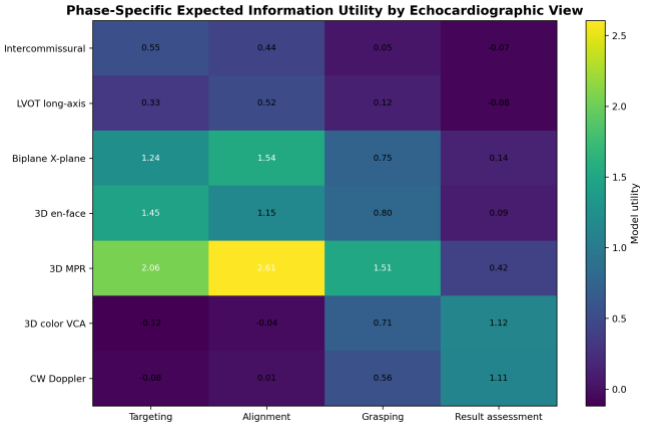


Figure 3. *Phase-specific expected utility of candidate echocardiographic views under the reference covariance state.*

The evolution of normalized posterior uncertainty (Figure 2)across the three-step reference acquisition sequence selected by the phase-aware policy. The sequence begins with 3D MPR, which primarily reduces uncertainty in coaptation geometry, followed by 3D color VCA, which provides additional information on residual mitral regurgitation, and CW Doppler, which substantially reduces uncertainty in transmitral gradient. Across the complete sequence, posterior standard deviations decreased from 2.60 to 0.520 mm for coaptation gap, from 1.40 to 0.459 mm for coaptation length, from 0.140 to 0.0489 cm^2 for residual 3D-VCA, and from 1.50 to 0.384 mmHg for transmitral gradient. The three acquisitions produced a cumulative information gain of 5.432 nats and satisfied the prespecified multidomain uncertainty guardrail. These reductions represent model-based posterior uncertainty rather than measured echocardiographic precision.

Figure 3 shows how the expected utility of each candidate echocardiographic acquisition varies across targeting, alignment, grasping, and result-assessment phases under the reference covariance state. The heatmap explains that view utility is phase-dependent rather than constant throughout the procedure. 3D MPR has the greatest utility during the early targeting and alignment phases and remains informative during grasping, consistent with its role in resolving coaptation geometry. In contrast, 3D colour VCA and CW Doppler become more valuable during result assessment, when uncertainty concerning residual MR and transmitral hemodynamics becomes more relevant. Biplane X-plane and 3D en-face imaging provide intermediate utility across the geometry-oriented phases. The phase-dependent redistribution of utility supports the model premise that the preferred acquisition should depend on both the remaining uncertainty and the current procedural phase rather than follow a universally fixed imaging sequence.

3. RESULTS

a) Mathematical Consistency and Reference Acquisition Sequence

All covariance matrices remained positive semidefinite across the evaluated acquisition sequences, and each observation produced a non-increasing posterior entropy relative to its corresponding predicted covariance, consistent with the specified linear-Gaussian update. In the reference computational scenario, the phase-aware policy selected 3D MPR as the first acquisition during the targeting phase. This acquisition reduced normalized coaptation-gap SD from 1.000 to 0.189 and normalized coaptation-length SD from 1.000 to 0.318, with an information gain of 2.868 nats. The second acquisition, 3D colour VCA, contributed an additional 1.309 nats and reduced residual 3D-VCA SD from 0.140 to 0.0746 cm^2. CW Doppler was selected third, contributing 1.255 nats and reducing transmitral-gradient SD from 1.50 to 0.384 mmHg.

The complete three-acquisition sequence yielded a cumulative information gain of 5.432 nats at a total acquisition penalty of 3.95 dimensionless model units. At the stopping point, posterior SDs were 0.520 mm for coaptation gap, 0.459 mm for coaptation length, 0.0489 cm^2 for residual 3D-VCA, and 0.384 mmHg for transmitral gradient, satisfying all four prespecified computational stopping guardrails. Posterior means in the reference synthetic state were 6.02 mm, 3.43 mm, 0.285 cm^2, and 3.47 mmHg, respectively. These values are outputs of the reference computational state and should not be interpreted as measurements from an individual patient.

b) Synthetic Policy Stress Testing

Across 600 synthetic scenarios, both the phase-aware and entropy-only policies reached the multidomain stopping guardrail with a median of three acquisitions. The prespecified fixed sequence required a median of six acquisitions, whereas the variance-greedy policy exhausted the eight-view computational horizon without satisfying the four-domain stopping criterion. Under the specified model assumptions, the phase-aware framework therefore required fewer acquisitions than the fixed sequence while satisfying the same multidomain uncertainty criterion. However, its median acquisition count was identical to that of entropy-only selection; consequently, these synthetic experiments do not establish superiority of phase-aware selection over every adaptive comparator. The 600 scenarios were generated solely for computational stress testing and do not represent patients, a synthetic clinical cohort, or estimates of population-level treatment effects.

Table 3. Synthetic policy stress tests summary (scenarios are not patients).

Policy	Median Views	Terminal Entropy	Weighted Uncertainty	VCA MAE	Gradient MAE	Completion Rate
Entropy	3.0	0.125	0.268	0.055	0.266	1.0
Fixed	6.0	−0.271	0.265	0.033	0.300	1.0
Full	3.0	0.156	0.269	0.038	0.321	1.0
No phase	3.0	0.147	0.293	0.057	0.292	1.0
Variance	8.0	0.331	0.463	0.115	0.184	0.0

c) Internal Mechanism Recovery Analysis

A separate experiment using 1,200 model-generated decision states evaluated whether alternative statistical representations could recover the stochastic action-selection mechanism specified by the full computational framework. The full phase-aware representation produced the lowest AIC (715.0) and BIC

(760.8), with an internal action agreement of 0.910. Corresponding AIC values were 4293.4 for entropy-only, 4580.6 for geometry-weighted selection, 9736.5 for clinical weighting without phase, and 10751.3 for cost-only selection.

These comparisons quantify recovery of the prespecified model-generated action mechanism rather than prediction of clinical decisions. Because the stochastic action labels were generated from the full computational framework itself, the AIC/BIC experiment constitutes an internal structural recovery analysis and does not provide evidence of clinical discrimination, calibration, external predictive validity, or patient-level benefit.

Table 4. Internal Model-Recovery Results

Model	k	AIC	BIC	Internal Action Agreement
Full phase-aware	9	715.0	760.8	0.910
Entropy only	2	4293.4	4303.6	0.737
Geometry weighted	3	4580.6	4595.9	0.736
Clinical weights no phase	5	9736.5	9761.9	0.201
Cost only	1	10751.3	10756.4	0.113

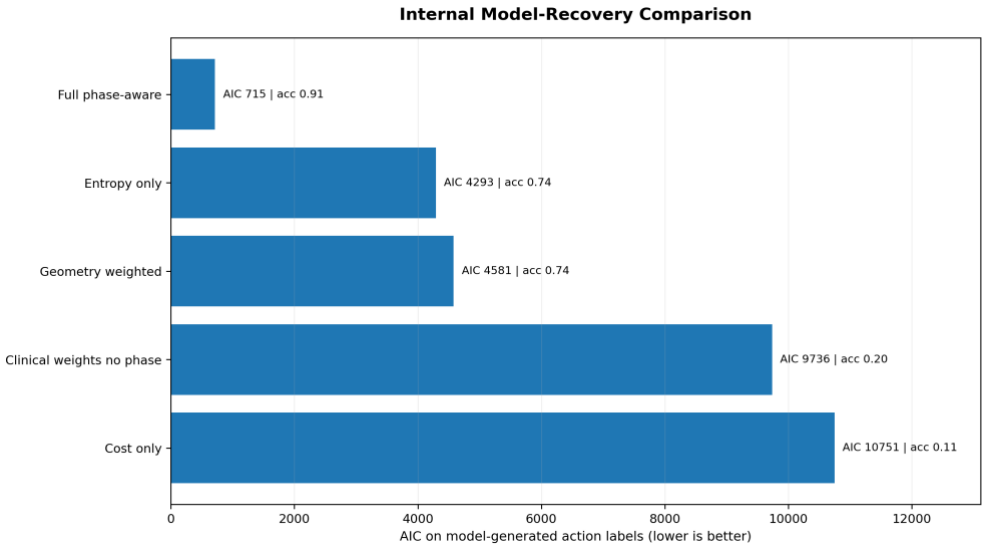


Figure 4. *Internal AIC comparison across five model representations using model-generated action labels; not clinical predictive performance.*

d) Global Sensitivity Analysis

Across 500 Latin-hypercube parameter sets, the coaptation-length stopping tolerance showed the strongest positive association with terminal posterior entropy (PRCC = 0.561), followed by the coaptation-gap stopping tolerance (PRCC = 0.287) and residual 3D-VCA stopping tolerance (PRCC = 0.250). Acquisition-cost weight and process-noise scaling showed smaller associations within the prespecified parameter ranges. These results indicate that residual model uncertainty was particularly sensitive to the stopping tolerances assigned to coaptation geometry and residual MR under the evaluated parameter space. Because the sampled quantities were provisional modelling parameters, the PRCC results characterize sensitivity of the computational model to its assumptions rather than biological or epidemiological associations in patients.

Table 5. Prespecified Structural Sensitivity Scenarios.

Scenario	Terminal Entropy	Weighted Uncertainty
Reference	0.156	0.292
High image noise	0.272	0.410
Low image noise	0.086	0.287
High acquisition penalty	0.125	0.288
Low acquisition penalty	0.156	0.292
Residual-MR priority	0.156	0.300
Geometry priority	0.125	0.272
Hemodynamic priority	0.125	0.187

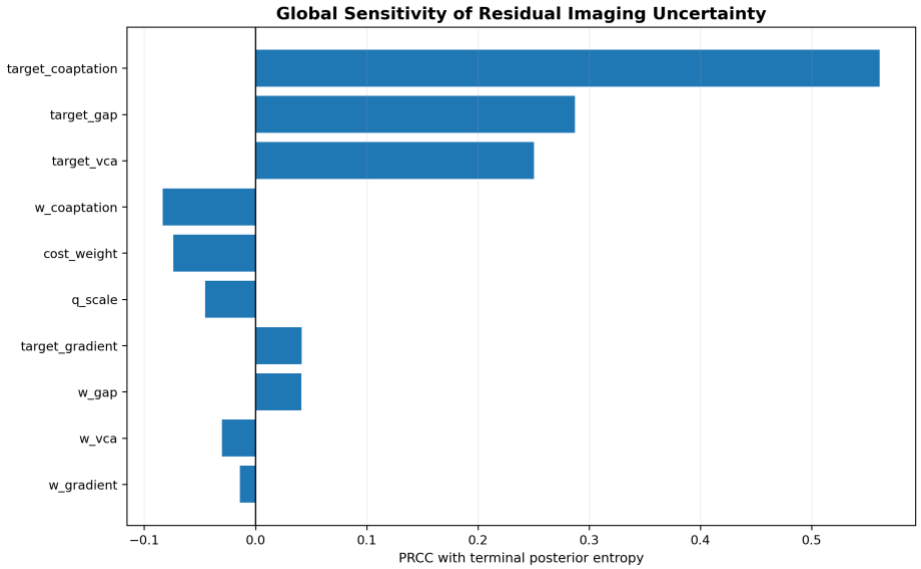


Figure 5. *PRCC sensitivity of terminal posterior entropy to provisional structural and stopping parameters.*

e) Covariance-Map Stability

Repeated application of individual view-specific covariance maps produced spectral radii close to one because each individual acquisition left one or more components of the four-dimensional latent state incompletely observed. In contrast, the complementary 3D-MPR to 3D-color-VCA to CW-Doppler acquisition cycle produced a spectral radius of 0.571. A spectral radius below one indicates local contraction of the corresponding covariance map around the evaluated reference state, supporting progressive reduction of the full modelled uncertainty state under repeated complementary acquisition. This result characterizes mathematical stability of the posterior covariance dynamics under the specified observation model and should not be interpreted as evidence of physiological stability or clinical outcome improvement.

4. DISCUSSION

This study developed and computationally evaluated an information-guided framework for sequential echocardiographic view selection during M-TEER. In the reference scenario, the phase-aware policy selected 3D MPR, followed by 3D color VCA and CW Doppler, satisfying all four prespecified uncertainty guardrails after three acquisitions. This sequence reflects the complementary roles of anatomical, residual-MR, and hemodynamic assessment in contemporary M-TEER imaging. Rather than treating echocardiographic views as interchangeable, the framework represents their contributions within a common latent state and quantifies how each acquisition reduces remaining uncertainty. The reference sequence accumulated 5.432 nats of information gain, with 3D MPR contributing the largest initial gain (2.868 nats), followed by 3D color VCA (1.309 nats) and CW Doppler (1.255 nats). This complementary behavior was also supported mathematically by the covariance analysis: repeated single-view maps had spectral radii close to one, whereas the complementary MPR-VCA-Doppler cycle produced a spectral radius of 0.571, indicating contraction of the multidomain covariance state under the specified linear-Gaussian model. These findings support the central modelling premise that acquisition value depends on the uncertainty remaining across multiple imaging domains.

Across 600 synthetic scenarios, both the phase-aware and entropy-only policies reached the stopping guardrail with a median of three acquisitions, compared with six for the prespecified fixed sequence, while the variance-greedy policy did not satisfy the four-domain criterion within the eight-view horizon. Thus, the simulations support uncertainty-adaptive acquisition relative to the tested fixed strategy but do not establish superiority of phase conditioning over entropy-only selection. Similarly, the internal model-recovery experiment showed the lowest AIC (715.0) and BIC (760.8) and an internal action agreement of 0.910 for the full phase-aware representation. Because the action labels were generated by the specified full model, these findings demonstrate internal structural recoverability rather than clinical predictive superiority. Sensitivity analysis further showed that terminal uncertainty was most strongly associated with coaptation-length stopping tolerance (PRCC = 0.561), followed by coaptation-gap (0.287) and residual 3D-VCA (0.250) tolerances. This dependence emphasizes that the current stopping thresholds are computational guardrails rather than empirically validated clinical precision criteria. The framework therefore differs from a conventional clinical prediction model: it does not predict mortality, procedural success, or treatment response, but instead formulates intraprocedural imaging as a sequential partially observed decision problem in which each acquisition updates the belief state and alters the value of subsequent information.

The present findings should be interpreted as a mathematical and computational proof-of-concept rather than evidence of clinical effectiveness. A computationally efficient acquisition sequence does not necessarily represent a clinically optimal sequence, and lower posterior uncertainty does not imply improved procedural or patient outcomes. Future validation should estimate observation noise, phase weights, acquisition costs, and stopping thresholds from time-stamped intraprocedural M-TEER data and compare model-guided recommendations with expert-directed imaging. If externally validated, the framework could provide a transparent decision-support layer for identifying which imaging domain remains uncertain and whether an additional acquisition is informative, while leaving procedural decisions under Heart Team control.

5. LIMITATIONS

a. No patient-level validation. The framework was evaluated computationally and has not been validated using real intraprocedural M-TEER data; therefore, the reported results should not be interpreted as clinical performance estimates.
b. Provisional model parameters. Observation matrices, measurement-noise covariances, phase weights, acquisition costs, and stopping thresholds are modelling assumptions rather than parameters estimated from patient data.
c. Simplified latent state. The model represents M-TEER imaging through four variables coaptation gap, coaptation length, residual 3D-VCA, and transmitral gradient and does not capture the full anatomical and hemodynamic complexity of the procedure.
d. Synthetic evaluation. The 600 stress-test scenarios, 1,200 model-generated decision states, and 500 Latin-hypercube parameter sets characterize internal model behavior and sensitivity rather than interpatient variation or external clinical validity.
e. Internal model recovery. The AIC/BIC experiment used model-generated action labels and therefore demonstrates structural recoverability rather than clinical discrimination, calibration, or predictive superiority.
f. Clinical calibration is required. Future studies should estimate model parameters from time-stamped M-TEER imaging and evaluate the model against expert-directed acquisition before prospective clinical use.

6. CONCLUSION

This study presents a mathematically testable model for information-guided echocardiographic view selection during M-TEER. By representing uncertainty in coaptation geometry, residual MR, and transmitral hemodynamics within a common belief state, the model provides a transparent mechanism for ranking candidate acquisitions according to their expected information value and for stopping acquisition once prespecified multidomain uncertainty guardrails are satisfied. Computational experiments demonstrated the internal consistency, sensitivity, and covariance stability of this sequential decision framework under the evaluated assumptions. These findings establish mathematical feasibility rather than clinical effectiveness and should not be interpreted as patient-level recommendations. Future work should estimate observation characteristics, procedural-phase parameters, acquisition costs, and stopping thresholds from real time-stamped M-TEER imaging and evaluate model-guided acquisition against expert-directed practice. Clinical calibration and external validation are required before the model can be considered for prospective decision support.

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